

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Winter Semester Examination – December - 2019**

Branch: B. Tech Mechanical

Sem.:- V

Subject with Code: - Theory of Machine -II (BTMEC504)

Marks: 60

Date:- 16/12/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately Assume it and should mention it clearly

(Marks)

Que.1.

(2×6=12)

A) A pulley is driven by a flat belt, angle of lap being 120° . The belt is 100 mm wide by 6 mm thick and density 1000 kg/m^3 . If coefficient of friction is 0.3 & maximum stress in belt is not exceed 2 MPa. Find greatest power which the belt can transmit & corresponding speed of belt.

Example 11.10 – S.S. Khosla – 350 Page

~~B) Explain centrifugal Tension in belt.~~

Que.2.

(2×6=12)

~~A) What do you understand by the term 'Interference' & 'undercutting' as applied to gears~~

B) A pinion of 20 involutes teeth and 125mm pitch circle diameter drives a rack. The addendum of both pinion and rack is 6.25mm. What is the least pressure angle which can be used to avoid interference? With this pressure angle, find the length of the arc of contact and the minimum numbers of teeth in contact at a time.

Que.3. Attempt the following

~~A) Explain compound gear train with neat sketch.~~

(6)

B) In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 rpm in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 rpm in the clockwise direction, what will be the speed of gear B.

(6)

Que. 4.

(2×6=12)

A) The turning moment diagram for a petrol engine is drawn to the following scales : Turning moment, 1 mm = 5 N-m Crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm². The rotating parts are equivalent to a mass of 36kg at a radius of gyration of 150mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m.

B) Explain the terms. i) Sensitiveness of governor. ii) Stability of governor.

iii) Isochronism. iv) Hunting of governor.

Que. 5.

(2×6=12)

A) The mass of the turbine rotor of a ship is 20 tonnes and has a radius of gyration of 0.6m. Its speed is 2000 r.p.m. The ship pitches 6° above and 6° below the horizontal position. The complete oscillation takes 30 seconds and the motion is simple harmonic. Determine the following. i) Maximum gyroscopic couple. ii) Maximum angular acceleration of the ship during pitching and iii) The direction in which the bow will tend to turn when rising, if the rotation of the rotor is clockwise when looking from the left.

B) The turbine rotor of ship has a mass of 2000kg and rotates at a speed of 3000 r.p.m. clockwise when looking from a stern. The radius of gyration of the rotor is 0.5m. Determine the gyroscopic couple and its effect upon the ship when the ship is steering to the right in curve of 100m radius at a speed of 16.1 knots (1 knot = 1855 m/h) calculate also the torque and its effects when the ship is pitching in simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 50 seconds and the total angular displacement between the two extreme positions of pitching is 12°. Find the maximum acceleration during pitching motion.

Que. 6. Solve the following.

A) Explain Rayleigh's method.

(6)

B) Explain and derive expression for critical or whirling speed of a shaft.

(6)

*****End Paper*****

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Q. 1. A.

Solution. Given : $\theta = 120^\circ = 120 \times \pi / 180 = 2.1 \text{ rad}$; $b = 100 \text{ mm} = 0.1 \text{ m}$; $t = 6 \text{ mm} = 0.006 \text{ m}$; $\rho = 1000 \text{ kg/m}^3$; $\mu = 0.3$; $\sigma = 2 \text{ MPa} = 2 \times 10^6 \text{ N/m}^2$

Speed of the belt for greatest power

We know that maximum tension in the belt,

$$T = \sigma \cdot b \cdot t = 2 \times 10^6 \times 0.1 \times 0.006 = 1200 \text{ N}$$

and mass of the belt per metre length,

$$\begin{aligned} m &= \text{Area} \times \text{length} \times \text{density} = b \cdot t \cdot l \cdot \rho \\ &= 0.1 \times 0.006 \times 1 \times 1000 = 0.6 \text{ kg/m} \end{aligned}$$

$\therefore$ Speed of the belt for greatest power,

$$v = \sqrt{\frac{T}{3m}} = \sqrt{\frac{1200}{3 \times 0.6}} = 25.82 \text{ m/s} \quad \text{Ans.}$$

Greatest power which the belt can transmit

We know that for maximum power to be transmitted, centrifugal tension,

$$T_C = T/3 = 1200/3 = 400 \text{ N}$$

and tension in the tight side of the belt,

$$T_1 = T - T_C = 1200 - 400 = 800 \text{ N}$$

Let

T_2 = Tension in the slack side of the belt.

We know that

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta = 0.3 \times 2.1 = 0.63$$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.63}{2.3} = 0.2739 \quad \text{or} \quad \frac{T_1}{T_2} = 1.88 \quad \dots (\text{Taking antilog of } 0.2739)$$

and

$$T_2 = \frac{T_1}{1.88} = \frac{800}{1.88} = 425.5 \text{ N}$$

$\therefore$ Greatest power which the belt can transmit,

$$P = (T_1 - T_2) v = (800 - 425.5) 25.82 = 9670 \text{ W} = 9.67 \text{ kW} \quad \text{Ans.}$$

ESE December 2019 TOM II (BTMEC504) Standard Solution

Q. 1. B.

11.16. Centrifugal Tension

Since the belt continuously runs over the pulleys, therefore, some centrifugal force is caused, whose effect is to increase the tension on both, tight as well as the slack sides. The tension caused by centrifugal force is called *centrifugal tension*. At lower belt speeds (less than 10 m/s), the centrifugal tension is very small, but at higher belt speeds (more than 10 m/s), its effect is considerable and thus should be taken into account.

Consider a small portion PQ of the belt subtending an angle $d\theta$ the centre of the pulley as shown in Fig. 11.17.

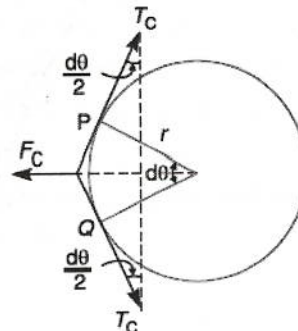


Fig. 11.17. Centrifugal tension.

- Let
- m = Mass of the belt per unit length in kg,
 - v = Linear velocity of the belt in m/s,
 - r = Radius of the pulley over which the belt runs in metres, and
 - T_c = Centrifugal tension acting tangentially at P and Q in newtons.

We know that length of the belt PQ

$$= r \cdot d\theta$$

and mass of the belt PQ

$$= m \cdot r \cdot d\theta$$

$\therefore$ Centrifugal force acting on the belt PQ ,

$$F_c = (m \cdot r \cdot d\theta) \frac{v^2}{r} = m \cdot d\theta \cdot v^2$$

The centrifugal tension T_c acting tangentially at P and Q keeps the belt in equilibrium.

Now resolving the forces (i.e. centrifugal force and centrifugal tension) horizontally and equating the same, we have

$$T_c \sin\left(\frac{d\theta}{2}\right) + T_c \sin\left(\frac{d\theta}{2}\right) = F_c = m \cdot d\theta \cdot v^2$$

Since the angle $d\theta$ is very small, therefore, putting $\sin\left(\frac{d\theta}{2}\right) = \frac{d\theta}{2}$, in the above expression,

$$2T_c \left(\frac{d\theta}{2}\right) = m \cdot d\theta \cdot v^2 \text{ or } T_c = m \cdot v^2$$

Notes : 1. When the centrifugal tension is taken into account, then total tension in the tight side,

$$T_{t1} = T_1 + T_c$$

and total tension in the slack side,

$$T_{t2} = T_2 + T_c$$

2. Power transmitted,

$$P = (T_{t1} - T_{t2}) v \quad \dots(\text{in watts})$$

$$= [(T_1 + T_c) - (T_2 + T_c)] v = (T_1 - T_2) v \quad \dots(\text{same as before})$$

Thus we see that centrifugal tension has no effect on the power transmitted.

3. The ratio of driving tensions may also be written as

$$2.3 \log \left(\frac{T_{t1} - T_c}{T_{t2} - T_c} \right) = \mu \cdot \theta$$

where

$$T_{t1} = \text{Maximum or total tension in the belt.}$$

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Q. 2. A.

12.19. Interference in Involute Gears

Fig. 12.13 shows a pinion with centre O_1 , in mesh with wheel or gear with centre O_2 . MN is the common tangent to the base circles and KL is the path of contact between the two mating teeth.

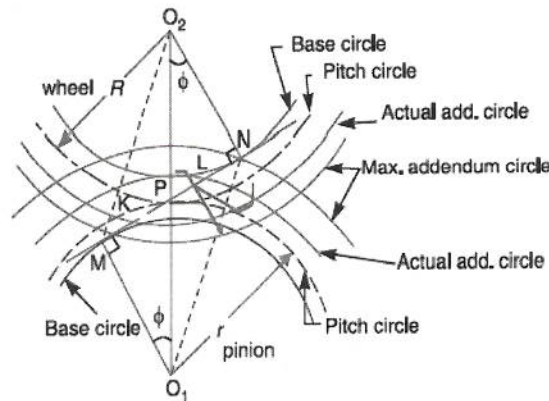


Fig. 12.13. Interference in involute gears.

A little consideration will show, that if the radius of the addendum circle of pinion is increased to O_1N , the point of contact L will move from L to N . When this radius is further increased, the point of contact L will be on the inside of base circle of wheel and not on the involute profile of tooth on wheel. The tip of tooth on the pinion will then undercut the tooth on the wheel at the root and remove part of the involute profile of tooth on the wheel. This effect is known as *interference*, and occurs when the teeth are being cut. In brief, *the phenomenon when the tip of tooth undercuts the root on its mating gear is known as interference.*

Similarly, if the radius of the addendum circle of the wheel increases beyond O_2M , then the tip of tooth on wheel will cause interference with the tooth on pinion. The points M and N are called *interference points*. Obviously, interference may be avoided if the path of contact does not extend beyond interference points. The limiting value of the radius of the addendum circle of the pinion is $*O_1N$ and of the wheel is O_2M .

From the above discussion, we conclude that the interference may only be avoided, if the point of contact between the two teeth is always on the involute profiles of both the teeth. In other

10.16

Undercutting

Figure 10.28(a) shows a pinion. A portion of its dedendum falls inside the base circle. The profile of the tooth inside the base circle is radial. If the addendum of the mating gear is more than the limiting value, it interferes with the dedendum of the pinion and the two gears are locked.

However, if a cutting rack having similar teeth is used to cut the teeth in the pinion, it will remove that portion of the pinion tooth which would have interfered with the gear as shown in Fig. 10.28(b). A gear having its material removed in this manner is said to be *undercut* and the process, *undercutting*. In a pinion with small number of teeth, no interference occurs. When the actual gear meshes with the undercut pinion, no interference occurs.

Undercutting will not take place if the teeth are designed to avoid interference.

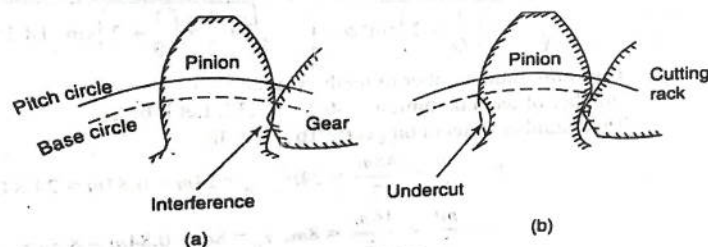


Fig. 10.28

ESE December 2019 TOM II (BTMEC504) Standard Solution

Q. 2. B.

Solution. Given : $T = 20$; $d = 125$ mm or $r = OP = 62.5$ mm ; $LH = 6.25$ mm

Least pressure angle to avoid interference

Let ϕ = Least pressure angle to avoid interference.

We know that for no interference, rack addendum,

$$LH = r \sin^2 \phi \quad \text{or} \quad \sin^2 \phi = \frac{LH}{r} = \frac{6.25}{62.5} = 0.1$$

$$\therefore \sin \phi = 0.3162 \quad \text{or} \quad \phi = 18.435^\circ \text{ Ans.}$$

Length of the arc of contact

We know that length of the path of contact,

$$\begin{aligned} KL &= \sqrt{(OK)^2 - (OL)^2} && \dots(\text{Refer Fig. 12.14}) \\ &= \sqrt{(OP + 6.25)^2 - (OP \cos \phi)^2} \\ &= \sqrt{(62.5 + 6.25)^2 - (62.5 \cos 18.435^\circ)^2} \\ &= \sqrt{4726.56 - 3515.62} = 34.8 \text{ mm} \end{aligned}$$

$\therefore$ Length of the arc of contact

$$= \frac{\text{Length of the path of contact}}{\cos \phi} = \frac{34.8}{\cos 18.435^\circ} = 36.68 \text{ mm Ans.}$$

Minimum number of teeth

We know that circular pitch,

$$p_c = \pi d / T = \pi \times 125 / 20 = 19.64 \text{ mm}$$

and the number of pairs of teeth in contact

$$= \frac{\text{Length of the arc of contact}}{\text{Circular pitch } (p_c)} = \frac{36.68}{19.64} = 1.87$$

$\therefore$ Minimum number of teeth in contact

$$= 2 \text{ or one pair Ans.}$$

Q. 3. A.

11.2

Compound Gear Train

When a series of gears are connected in such a way that two or more gears rotate about an axis with the same angular velocity, it is known as compound gear train. In this type, some of the intermediate shafts, i.e., other than the input and the output shafts, carry more than one gear as shown in Fig. 11.2.

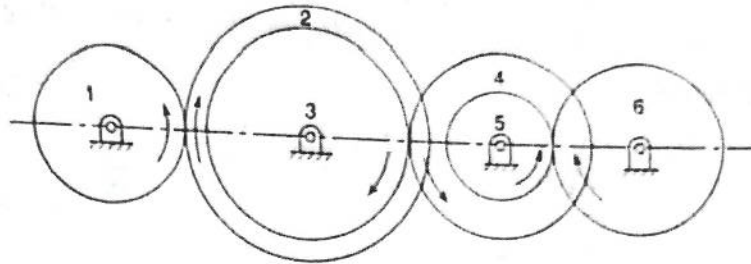


Fig. 11.2

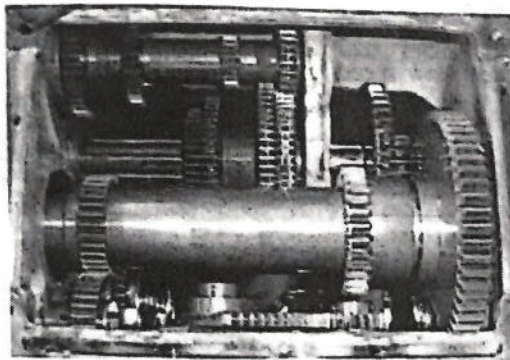


Illustration 11.1: Gear box of a lathe consisting of compound gears

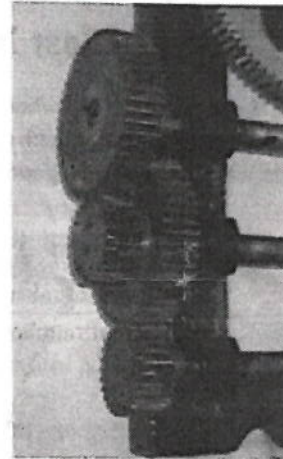


Illustration 11.2: A compound gear train

If the gear 1 is the driver then

$$\frac{N_2}{N_1} = \frac{T_1}{T_2}, \quad \frac{N_4}{N_3} = \frac{T_3}{T_4} \quad \text{and} \quad \frac{N_6}{N_5} = \frac{T_5}{T_6}$$

or

$$\frac{N_2}{N_1} \times \frac{N_4}{N_3} \times \frac{N_6}{N_5} = \frac{T_1}{T_2} \times \frac{T_3}{T_4} \times \frac{T_5}{T_6}$$

or

$$\frac{N_2}{N_1} \times \frac{N_4}{N_2} \times \frac{N_6}{N_4} = \frac{T_1}{T_2} \times \frac{T_3}{T_4} \times \frac{T_5}{T_6}$$

$$\frac{N_6}{N_1} = \frac{T_1}{T_2} \times \frac{T_3}{T_4} \times \frac{T_5}{T_6}$$

(11.2)

or

$$\text{Train value} = \frac{\text{Product of number of teeth on driving gears}}{\text{Product of number of teeth on driven gears}}$$

ESE December 2019 TOM II (BTMEC504) Standard Solution

Q. 3. B.

Example 13.4. In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 r.p.m. in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 r.p.m. in the clockwise direction, what will be the speed of gear B?

Solution. Given : $T_A = 36$; $T_B = 45$; $N_C = 150$ r.p.m. (anticlockwise)

The gear train is shown in Fig. 13.7.

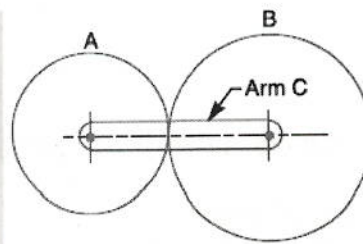


Fig. 13.7

1. Tabular method

First of all prepare the table of motions as given below :

Table 13.2. Table of motions.

Step No.	Conditions of motion	Revolutions of elements		
		Arm C	Gear A	Gear B
1.	Arm fixed-gear A rotates through + 1 revolution (i.e. 1 rev. anticlockwise)	0	+ 1	$-\frac{T_A}{T_B}$
2.	Arm fixed-gear A rotates through + x revolutions	0	+ x	$-x \times \frac{T_A}{T_B}$
3.	Add + y revolutions to all elements	+ y	+ y	+ y
4.	Total motion	+ y	x + y	$y - x \times \frac{T_A}{T_B}$

Speed of gear B when gear A is fixed

Since the speed of arm is 150 r.p.m. anticlockwise, therefore from the fourth row of the table,

$$y = + 150 \text{ r.p.m.}$$

Also the gear A is fixed, therefore

$$x + y = 0 \quad \text{or} \quad x = -y = -150 \text{ r.p.m.}$$

$$\therefore \text{Speed of gear B, } N_B = y - x \times \frac{T_A}{T_B} = 150 + 150 \times \frac{36}{45} = + 270 \text{ r.p.m.}$$

$$= 270 \text{ r.p.m. (anticlockwise) Ans.}$$

Speed of gear B when gear A makes 300 r.p.m. clockwise

Since the gear A makes 300 r.p.m. clockwise, therefore from the fourth row of the table,

$$x + y = -300 \quad \text{or} \quad x = -300 - y = -300 - 150 = -450 \text{ r.p.m.}$$

$\therefore$ Speed of gear B,

$$N_B = y - x \times \frac{T_A}{T_B} = 150 + 450 \times \frac{36}{45} = + 510 \text{ r.p.m.}$$

$$= 510 \text{ r.p.m. (anticlockwise) Ans.}$$

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2. Algebraic method

Let N_A = Speed of gear A.
 N_B = Speed of gear B, and
 N_C = Speed of arm C.

Assuming the arm C to be fixed, speed of gear A relative to arm C

$$= N_A - N_C$$

and speed of gear B relative to arm C = $N_B - N_C$

Since the gears A and B revolve in opposite directions, therefore

$$\frac{N_B - N_C}{N_A - N_C} = -\frac{T_A}{T_B} \quad \dots(i)$$

Speed of gear B when gear A is fixed

When gear A is fixed, the arm rotates at 150 r.p.m. in the anticlockwise direction, i.e.

$$N_A = 0, \quad \text{and} \quad N_C = +150 \text{ r.p.m.}$$

$$\therefore \frac{N_B - 150}{0 - 150} = -\frac{36}{45} = -0.8 \quad \dots[\text{From equation (i)}]$$

or $N_B = -150 \times -0.8 + 150 = 120 + 150 = 270 \text{ r.p.m. Ans.}$

Speed of gear B when gear A makes 300 r.p.m. clockwise

Since the gear A makes 300 r.p.m. clockwise, therefore

$$N_A = -300 \text{ r.p.m.}$$

$$\therefore \frac{N_B - 150}{-300 - 150} = -\frac{36}{45} = -0.8$$

or $N_B = -450 \times -0.8 + 150 = 360 + 150 = 510 \text{ r.p.m. Ans.}$

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Q. 4. A.

Example 16.4. The turning moment diagram for a petrol engine is drawn to the following scales : Turning moment, 1 mm = 5 N-m ; crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm². The rotating parts are equivalent to a mass of 36 kg at a radius of gyration of 150 mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m.

Solution. Given : $m = 36 \text{ kg}$; $k = 150 \text{ mm} = 0.15 \text{ m}$; $N = 1800 \text{ r.p.m.}$ or $\omega = 2\pi \times 1800/60 = 188.52 \text{ rad/s}$

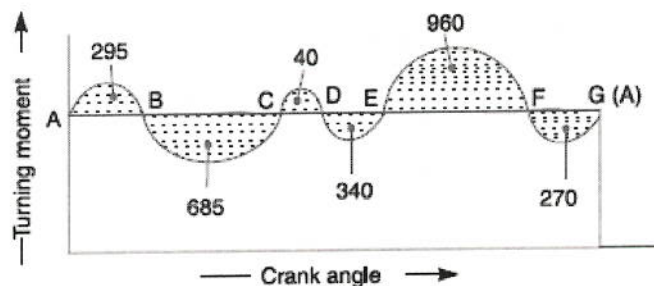


Fig. 16.6

The turning moment diagram is shown in Fig. 16.6.

Since the turning moment scale is 1 mm = 5 N-m and crank angle scale is 1 mm = 1° = $\pi/180$ rad, therefore,

1 mm² on turning moment diagram

$$= 5 \times \frac{\pi}{180} = \frac{\pi}{36} \text{ N-m}$$

Let the total energy at $A = E$, then referring to Fig. 16.6,

$$\text{Energy at } B = E + 295$$

... (Maximum energy)

$$\text{Energy at } C = E + 295 - 685 = E - 390$$

$$\text{Energy at } D = E - 390 + 40 = E - 350$$

$$\text{Energy at } E = E - 350 - 340 = E - 690 \text{ ... (Minimum energy)}$$

$$\text{Energy at } F = E - 690 + 960 = E + 270$$

$$\text{Energy at } G = E + 270 - 270 = E = \text{Energy at } A$$

We know that maximum fluctuation of energy,

$$\Delta E = \text{Maximum energy} - \text{Minimum energy}$$

$$= (E + 295) - (E - 690) = 985 \text{ mm}^2$$

$$= 985 \times \frac{\pi}{36} = 86 \text{ N-m} = 86 \text{ J}$$

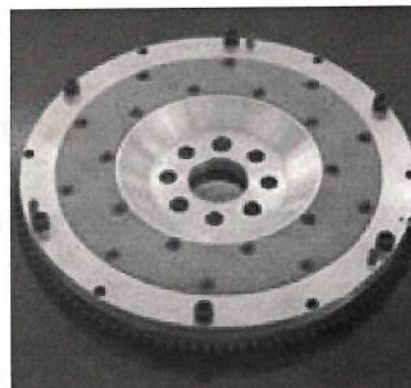
Let

C_s = Coefficient of fluctuation of speed.

We know that maximum fluctuation of energy (ΔE),

$$86 = m \cdot k^2 \omega^2 \cdot C_s = 36 \times (0.15)^2 \times (188.52)^2 C_s = 28\,787 C_s$$

$$\therefore C_s = 86 / 28\,787 = 0.003 \text{ or } 0.3\% \text{ Ans.}$$



Flywheel of an electric motor.

ESE December 2019 TOM II (BTMEC504) Standard Solution

Q. 4. B.

18.12. Sensitiveness of Governors

Consider two governors A and B running at the same speed. When this speed increases or decreases by a certain amount, the lift of the sleeve of governor A is greater than the lift of the sleeve of governor B . It is then said that the governor A is more sensitive than the governor B .

In general, the greater the lift of the sleeve corresponding to a given fractional change in speed, the greater is the sensitiveness of the governor. It may also be stated in another way that for a given lift of the sleeve, the sensitiveness of the governor increases as the speed range decreases. This definition of sensitiveness may be quite satisfactory when the governor is considered as an independent mechanism. But when the governor is fitted to an engine, the practical requirement is simply that the change of equilibrium speed from the full load to the no load position of the sleeve should be as small a fraction as possible of the mean equilibrium speed. The actual displacement of the sleeve is immaterial, provided that it is sufficient to change the energy supplied to the engine by the required amount. For this reason, the sensitiveness is defined as the *ratio of the difference between the maximum and minimum equilibrium speeds to the mean equilibrium speed*.

Let N_1 = Minimum equilibrium speed,

N_2 = Maximum equilibrium speed, and

$$N = \text{Mean equilibrium speed} = \frac{N_1 + N_2}{2}.$$

$\therefore$ Sensitiveness of the governor

$$= \frac{N_2 - N_1}{N} = \frac{2(N_2 - N_1)}{N_1 + N_2}$$

$$= \frac{2(\omega_2 - \omega_1)}{\omega_1 + \omega_2} \quad \dots \text{(In terms of angular speeds)}$$

18.13. Stability of Governors

A governor is said to be *stable* when for every speed within the working range there is a definite configuration *i.e.* there is only one radius of rotation of the governor balls at which the governor is in equilibrium. For a stable governor, if the equilibrium speed increases, the radius of governor balls must also increase.

Note : A governor is said to be unstable, if the radius of rotation decreases as the speed increases.

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18.14. Isochronous Governors

A governor is said to be *isochronous* when the equilibrium speed is constant (i.e. range of speed is zero) for all radii of rotation of the balls within the working range, neglecting friction. The isochronism is the stage of infinite sensitivity.

Let us consider the case of a Porter governor running at speeds N_1 and N_2 r.p.m. We have discussed in Art. 18.6 that

$$(N_1)^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{895}{h_1} \quad \dots (i)$$

and
$$(N_2)^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{895}{h_2} \quad \dots (ii)$$

For isochronism, range of speed should be zero i.e. $N_2 - N_1 = 0$ or $N_2 = N_1$. Therefore from equations (i) and (ii), $h_1 = h_2$, which is impossible in case of a Porter governor. Hence a Porter governor cannot be isochronous.

Now consider the case of a Hartnell governor running at speeds N_1 and N_2 r.p.m. We have discussed in Art. 18.8 that

$$M \cdot g + S_1 = 2 F_{C1} \times \frac{x}{y} = 2 \times m \left(\frac{2\pi N_1}{60} \right)^2 r_1 \times \frac{x}{y} \quad \dots (iii)$$

and
$$M \cdot g + S_2 = 2 F_{C2} \times \frac{x}{y} = 2 \times m \left(\frac{2\pi N_2}{60} \right)^2 r_2 \times \frac{x}{y} \quad \dots (iv)$$

For isochronism, $N_2 = N_1$. Therefore from equations (iii) and (iv),

$$\frac{M \cdot g + S_1}{M \cdot g + S_2} = \frac{r_1}{r_2}$$

Note : The isochronous governor is not of practical use because the sleeve will move to one of its extreme positions immediately the speed deviates from the isochronous speed.



18.15. Hunting

A governor is said to be *hunt* if the speed of the engine fluctuates continuously above and below the mean speed. This is caused by a too sensitive governor which changes the fuel supply by a large amount when a small change in the speed of rotation takes place. For example, when the load on the engine increases, the engine speed decreases and, if the governor is very sensitive, the governor sleeve immediately falls to its lowest position. This will result in the opening of the control valve wide which will supply the fuel to the engine in excess of its requirement so that the engine speed rapidly increases again and the governor sleeve rises to its highest position. Due to this movement of the sleeve, the control valve will cut off the fuel supply to the engine and thus the engine speed begins to fall once again. This cycle is repeated indefinitely.



A forklift is used to carry small loads from one place to the other inside a factory.

Such a governor may admit either the maximum or the minimum amount of fuel. The effect of this will be to cause wide fluctuations in the engine speed or in other words, the engine will hunt.

ESE December 2019 TOM II (BTMEC504) Standard Solution

Q. 5. A.

Example 14.7. The mass of the turbine rotor of a ship is 20 tonnes and has a radius of gyration of 0.60 m. Its speed is 2000 r.p.m. The ship pitches 6° above and 6° below the horizontal position. A complete oscillation takes 30 seconds and the motion is simple harmonic. Determine the following:

1. Maximum gyroscopic couple, 2. Maximum angular acceleration of the ship during pitching, and 3. The direction in which the bow will tend to turn when rising, if the rotation of the rotor is clockwise when looking from the left.

Solution. Given : $m = 20 \text{ t} = 20\,000 \text{ kg}$; $k = 0.6 \text{ m}$; $N = 2000 \text{ r.p.m.}$ or $\omega = 2\pi \times 2000/60 = 209.5 \text{ rad/s}$; $\phi = 6^\circ = 6 \times \pi/180 = 0.105 \text{ rad}$; $t_p = 30 \text{ s}$

1. Maximum gyroscopic couple

We know that mass moment of inertia of the rotor,

$$I = m \cdot k^2 = 20\,000 (0.6)^2 = 7200 \text{ kg-m}^2$$

and angular velocity of the simple harmonic motion,

$$\omega_1 = 2\pi / t_p = 2\pi/30 = 0.21 \text{ rad/s}$$

$\therefore$ Maximum angular velocity of precession,

$$\omega_{p_{\max}} = \phi \cdot \omega_1 = 0.105 \times 0.21 = 0.022 \text{ rad/s}$$

We know that maximum gyroscopic couple,

$$\begin{aligned} C_{\max} &= I \omega \omega_{p_{\max}} = 7200 \times 209.5 \times 0.022 = 33\,185 \text{ N-m} \\ &= 33.185 \text{ kN-m Ans.} \end{aligned}$$

2. Maximum angular acceleration during pitching

We know that maximum angular acceleration during pitching

$$= \phi(\omega_1)^2 = 0.105 (0.21)^2 = 0.0046 \text{ rad/s}^2$$

3. Direction in which the bow will tend to turn when rising

We have discussed in Art. 14.7, that when the rotation of the rotor is clockwise when looking from the left (*i.e.* rear end or stern) and when the bow is rising (*i.e.* pitching is upward), then the reactive gyroscopic couple acts in the clockwise direction which tends to turn the bow towards right (*i.e.* towards star-board). Ans.

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Q. 5. B.

Example 14.9. The turbine rotor of a ship has a mass of 2000 kg and rotates at a speed of 3000 r.p.m. clockwise when looking from a stern. The radius of gyration of the rotor is 0.5 m.

Determine the gyroscopic couple and its effects upon the ship when the ship is steering to the right in a curve of 100 m radius at a speed of 16.1 knots (1 knot = 1855 m/hr).

Calculate also the torque and its effects when the ship is pitching in simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 50 seconds and the total angular displacement between the two extreme positions of pitching is 12° . Find the maximum acceleration during pitching motion.

Solution. Given : $m = 2000$ kg ; $N = 3000$ r.p.m. or $\omega = 2\pi \times 3000/60 = 314.2$ rad/s ;
 $k = 0.5$ m ; $R = 100$ m ; $v = 16.1$ knots = $16.1 \times 1855 / 3600 = 8.3$ m/s

Gyroscopic couple

We know that mass moment of inertia of the rotor,

$$I = m k^2 = 2000 (0.5)^2 = 500 \text{ kg-m}^2$$

and angular velocity of precession,

$$\omega_p = v/R = 8.3/100 = 0.083 \text{ rad/s}$$

$\therefore$ Gyroscopic couple,

$$C = I \omega \omega_p = 500 \times 314.2 \times 0.083 = 13\,040 \text{ N-m} = 13.04 \text{ kN-m}$$

We have discussed in Art. 14.6, that when the rotor rotates clockwise when looking from a stern and the ship steers to the right, the effect of the reactive gyroscopic couple is to raise the stern and lower the bow. **Ans.**

Torque during pitching

Given : $t_p = 50$ s ; $2\phi = 12^\circ$ or $\phi = 6^\circ \times \pi/180 = 0.105$ rad

We know that angular velocity of simple harmonic motion,

$$\omega_1 = 2\pi/t_p = 2\pi/50 = 0.1257 \text{ rad/s}$$

and maximum angular velocity of precession,

$$\omega_{p_{\max}} = \phi \omega_1 = 0.105 \times 0.1257 = 0.0132 \text{ rad/s}$$

$\therefore$ Torque or maximum gyroscopic couple during pitching,

$$C_{\max} = I \omega \omega_{p_{\max}} = 500 \times 314.2 \times 0.0132 = 2074 \text{ N-m Ans.}$$

We have discussed in Art. 14.7, that when the pitching is downwards, the effect of the reactive gyroscopic couple is to turn the ship towards port side.

Maximum acceleration during pitching

We know that maximum acceleration during pitching

$$\alpha_{\max} = \phi (\omega_1)^2 = 0.105 (0.1257)^2 = 0.00166 \text{ rad/s}^2 \text{ Ans.}$$

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Q. 6. A.

3. Rayleigh's method

In this method, the maximum kinetic energy at the mean position is equal to the maximum potential energy (or strain energy) at the extreme position. Assuming the motion executed by the vibration to be simple harmonic, then

$$x = X \sin \omega t \quad \dots (i)$$

where

x = Displacement of the body from the mean position after time t seconds, and

X = Maximum displacement from mean position to extreme position.

Now, differentiating equation (i), we have

$$\frac{dx}{dt} = \omega \times X \cos \omega t$$

Since at the mean position, $t = 0$, therefore maximum velocity at the mean position,

$$v = \frac{dx}{dt} = \omega X$$

$\therefore$ Maximum kinetic energy at mean position

$$= \frac{1}{2} \times m \cdot v^2 = \frac{1}{2} \times m \cdot \omega^2 X^2 \quad \dots (ii)$$

and maximum potential energy at the extreme position

$$= \left(\frac{0 + s \cdot X}{2} \right) X = \frac{1}{2} \times s \cdot X^2 \quad \dots (iii)$$

Equating equations (ii) and (iii),

$$\frac{1}{2} \times m \cdot \omega^2 X^2 = \frac{1}{2} \times s \cdot X^2 \quad \text{or} \quad \omega^2 = \frac{s}{m}, \text{ and } \omega = \sqrt{\frac{s}{m}}$$

$$\therefore \text{ Time period, } t_p = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{s}{m}} \quad \dots (\text{Same as before})$$

$$\text{and natural frequency, } f_n = \frac{1}{t_p} = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{s}{m}} \quad \dots (\text{Same as before})$$

Note : In all the above expressions, ω is known as natural circular frequency and is generally denoted by ω_n

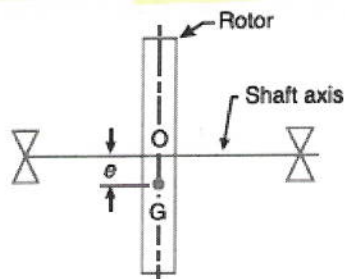
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Q. 6. B.

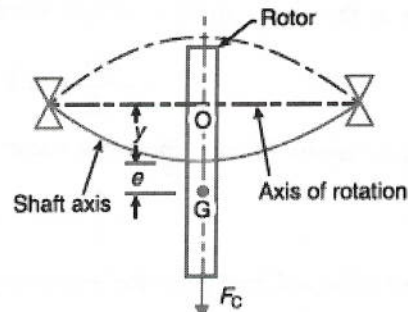
23.12. Critical or Whirling Speed of a Shaft

In actual practice, a rotating shaft carries different mountings and accessories in the form of gears, pulleys, etc. When the gears or pulleys are put on the shaft, the centre of gravity of the pulley or gear does not coincide with the centre line of the bearings or with the axis of the shaft, when the shaft is stationary. This means that the centre of gravity of the pulley or gear is at a certain distance from the axis of rotation and due to this, the shaft is subjected to centrifugal force. This force will bent the shaft which will further increase the distance of centre of gravity of the pulley or gear from the axis of rotation. This correspondingly increases the value of centrifugal force, which further increases the distance of centre of gravity from the axis of rotation. This effect is cumulative and ultimately the shaft fails. The bending of shaft not only depends upon the value of eccentricity (distance between centre of gravity of the pulley and the axis of rotation) but also depends upon the speed at which the shaft rotates.

The speed at which the shaft runs so that the additional deflection of the shaft from the axis of rotation becomes infinite, is known as *critical or whirling speed*.



(a) When shaft is stationary.



(b) When shaft is rotating.

Fig. 23.14. Critical or whirling speed of a shaft.

Consider a shaft of negligible mass carrying a rotor, as shown in Fig. 23.14 (a). The point O is on the shaft axis and G is the centre of gravity of the rotor. When the shaft is stationary, the centre line of the bearing and the axis of the shaft coincides. Fig. 23.14 (b) shows the shaft when rotating about the axis of rotation at a uniform speed of ω rad/s.

Let

m = Mass of the rotor,

e = Initial distance of centre of gravity of the rotor from the centre line of the bearing or shaft axis, when the shaft is stationary,

y = Additional deflection of centre of gravity of the rotor when the shaft starts rotating at ω rad/s, and

s = Stiffness of the shaft i.e. the load required per unit deflection of the shaft.

Since the shaft is rotating at ω rad/s, therefore centrifugal force acting radially outwards through G causing the shaft to deflect is given by

$$F_C = m \cdot \omega^2 (y + e)$$

The shaft behaves like a spring. Therefore the force resisting the deflection y ,

$$= s \cdot y$$

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For the equilibrium position,

$$m.\omega^2(y+e) = s.y$$

or $m.\omega^2.y + m.\omega^2.e = s.y$ or $y(s - m.\omega^2) = m.\omega^2.e$

$$\therefore y = \frac{m.\omega^2.e}{s - m.\omega^2} = \frac{\omega^2.e}{s/m - \omega^2} \quad \dots (i)$$

We know that circular frequency,

$$\omega_n = \sqrt{\frac{s}{m}} \quad \text{or} \quad y = \frac{\omega^2.e}{(\omega_n)^2 - \omega^2} \quad \dots [\text{From equation (i)}]$$

A little consideration will show that when $\omega > \omega_n$, the value of y will be negative and the shaft deflects in the opposite direction as shown dotted in Fig 23.14 (b).

In order to have the value of y always positive, both *plus* and *minus* signs are taken.

$$\therefore y = \pm \frac{\omega^2.e}{(\omega_n)^2 - \omega^2} = \frac{\pm e}{\left(\frac{\omega_n}{\omega}\right)^2 - 1} = \frac{\pm e}{\left(\frac{\omega_c}{\omega}\right)^2 - 1} \quad \dots (\text{Substituting } \omega_n = \omega_c)$$

We see from the above expression that when $\omega_n = \omega_c$, the value of y becomes infinite. Therefore ω_c is the critical or whirling speed.

$\therefore$ Critical or whirling speed,

$$\omega_c = \omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{g}{\delta}} \text{ Hz} \quad \dots \left(\because \delta = \frac{m.g}{s} \right)$$

If N_c is the critical or whirling speed in r.p.s., then

$$2\pi N_c = \sqrt{\frac{g}{\delta}} \quad \text{or} \quad N_c = \frac{1}{2\pi} \sqrt{\frac{g}{\delta}} = \frac{0.4985}{\sqrt{\delta}} \text{ r.p.s.}$$

where

δ = Static deflection of the shaft in metres.

Hence the critical or whirling speed is the same as the natural frequency of transverse vibration but its unit will be revolutions per second.